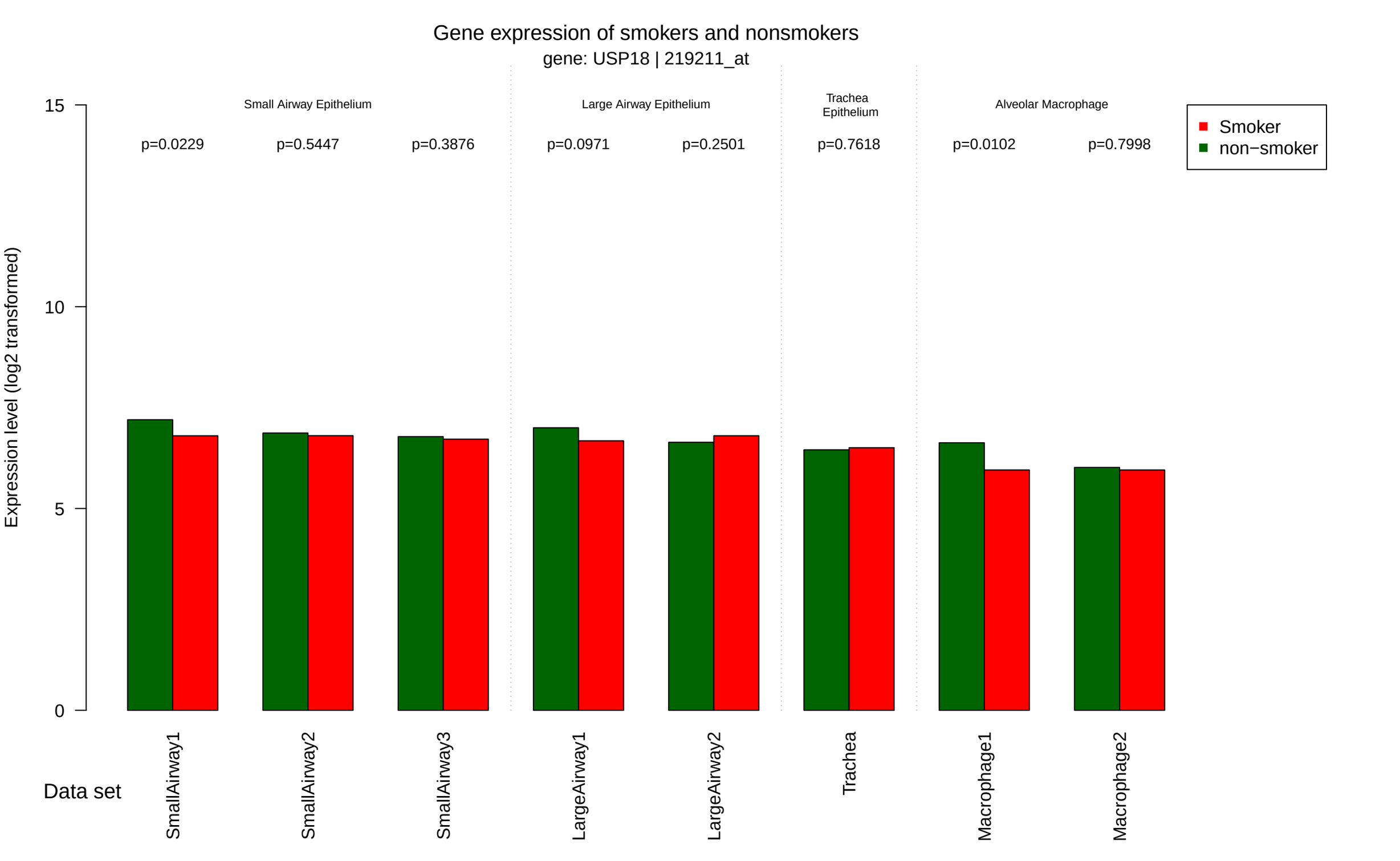
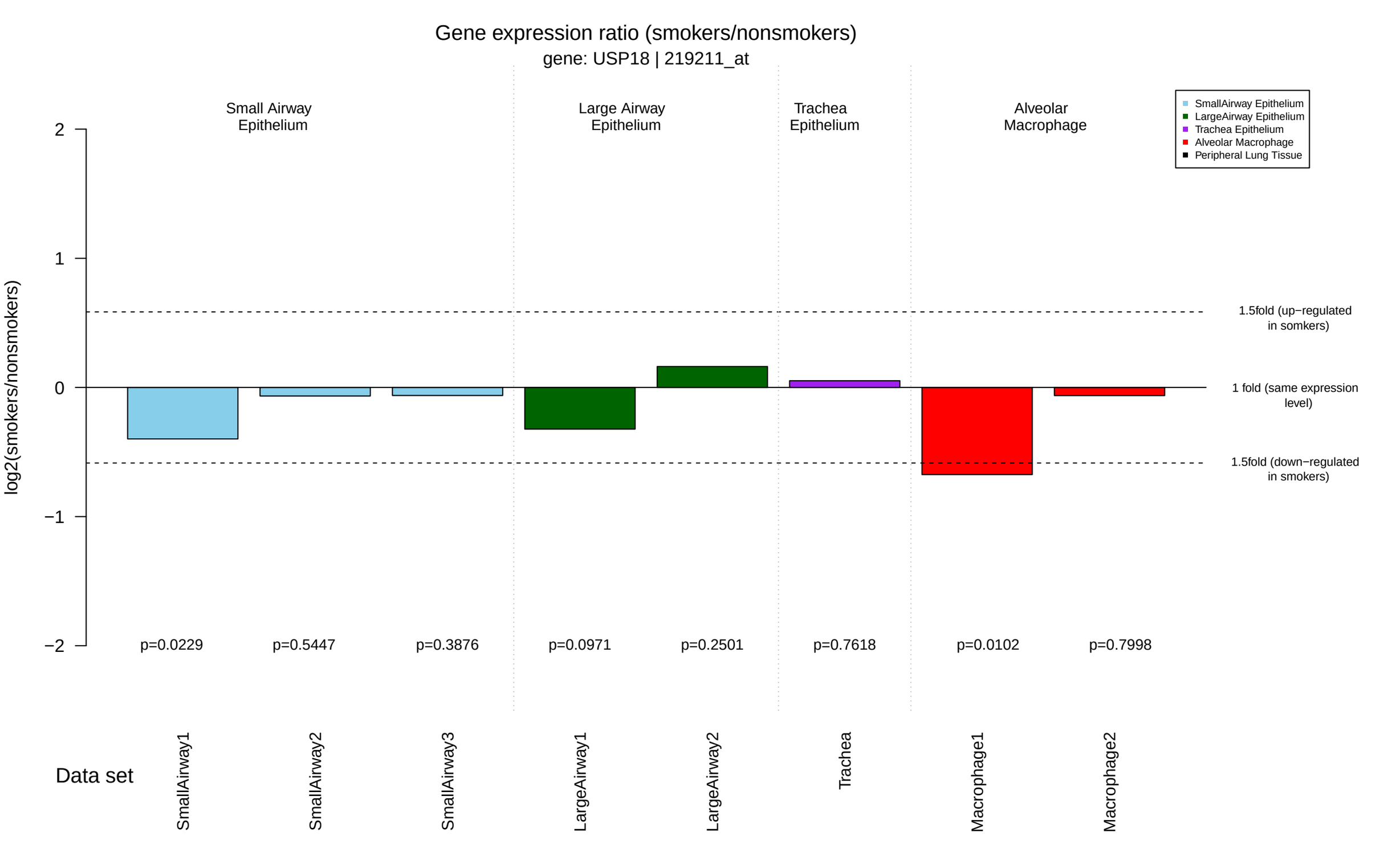
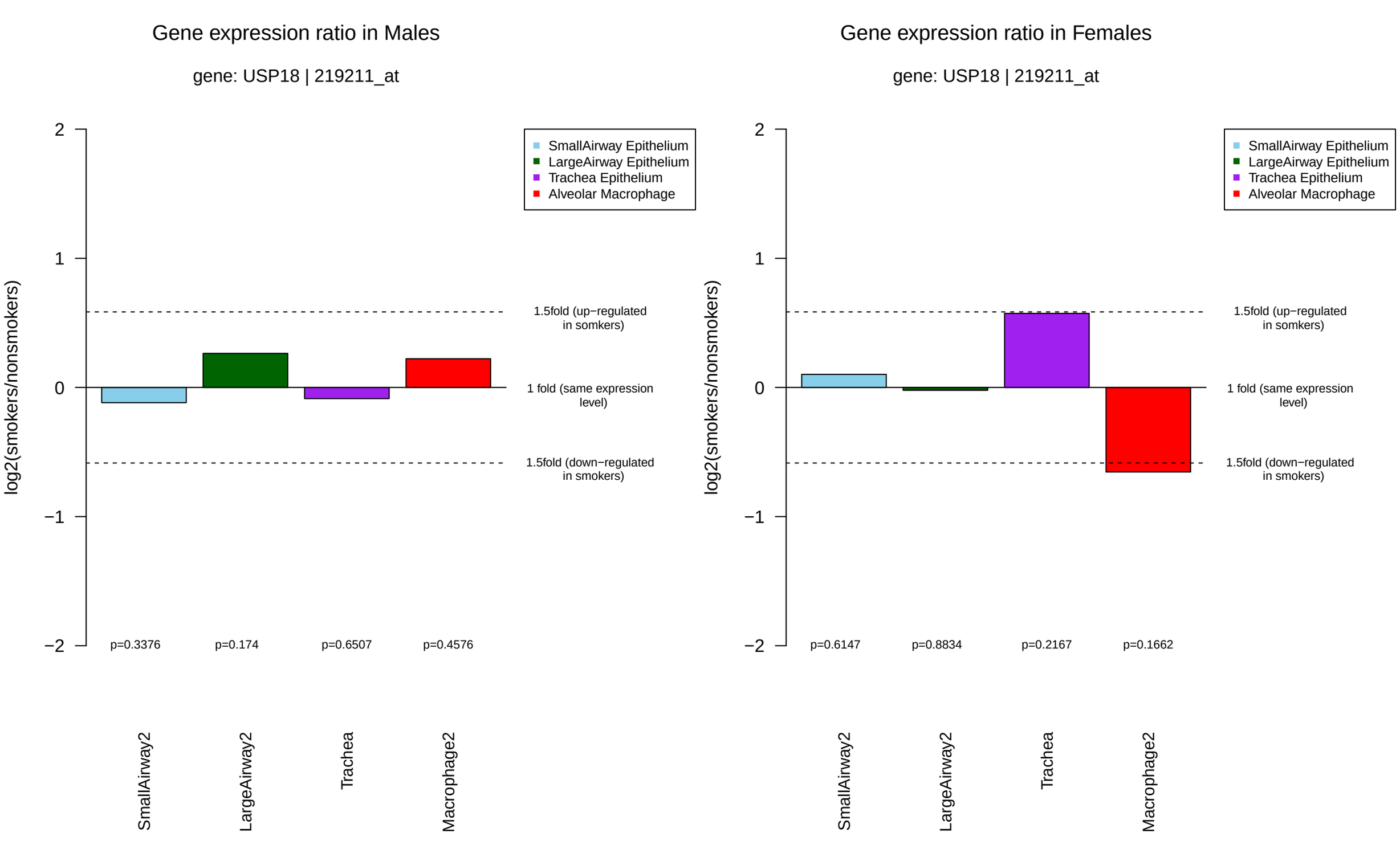
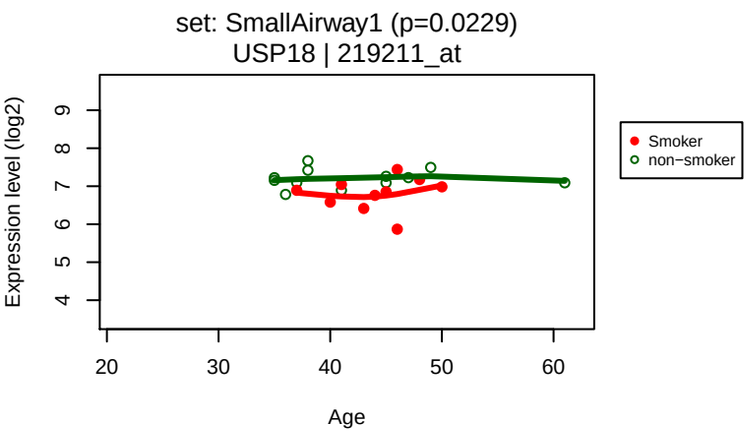
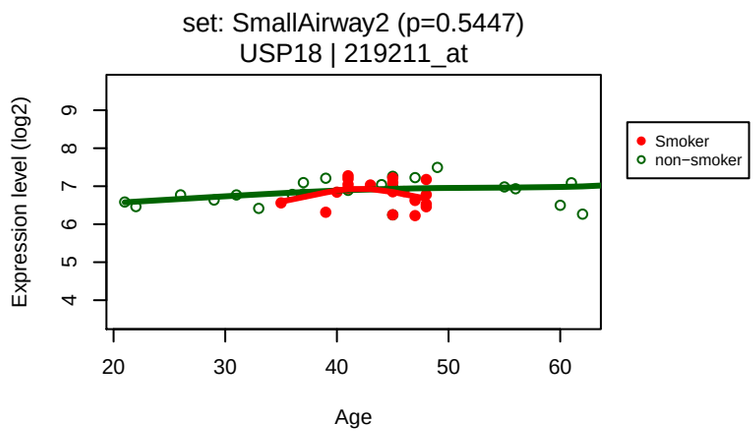
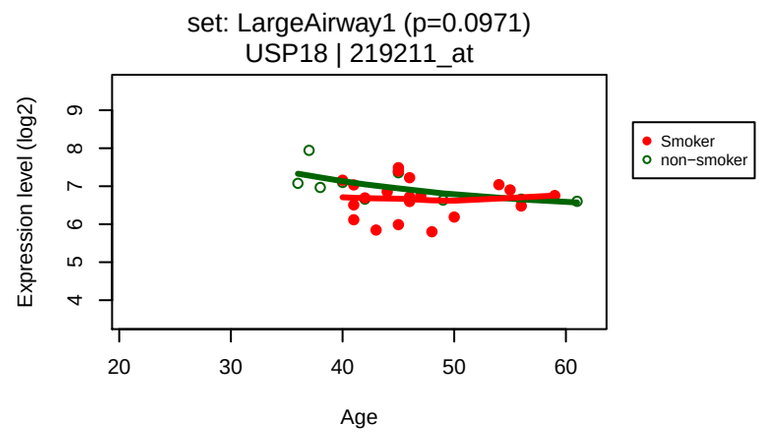

Gene expression of smokers and nonsmokers
gene: USP18 | 219211_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0229
p=0.5447
p=0.3876
p=0.0971
p=0.2501
p=0.7618
p=0.0102
p=0.7998
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

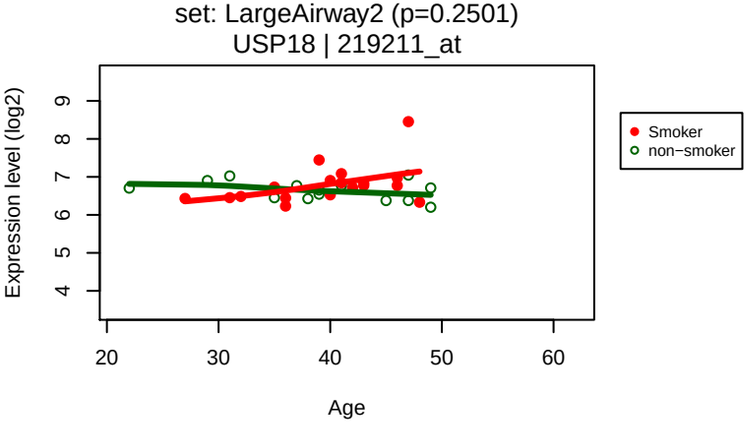

Gene expression ratio (smokers/nonsmokers)
gene: USP18 | 219211_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0229
p=0.5447
p=0.3876
p=0.0971
p=0.2501
p=0.7618
p=0.0102
p=0.7998
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

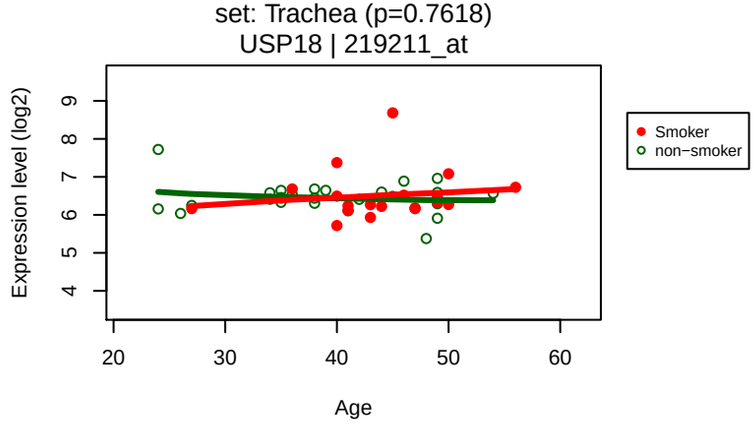

Gene expression ratio in Males
gene: USP18 | 219211_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3376
p=0.174
p=0.6507
p=0.4576
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP18 | 219211_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6147
p=0.8834
p=0.2167
p=0.1662
SmallAirway2
LargeAirway2
Trachea
Macrophage2

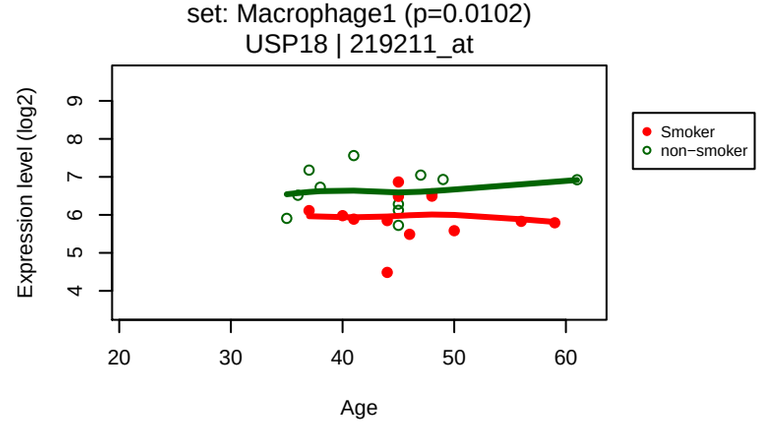

set: SmallAirway1 (p=0.0229)
USP18 | 219211_at
Expression level (log2)
Age
Smoker
non−smoker

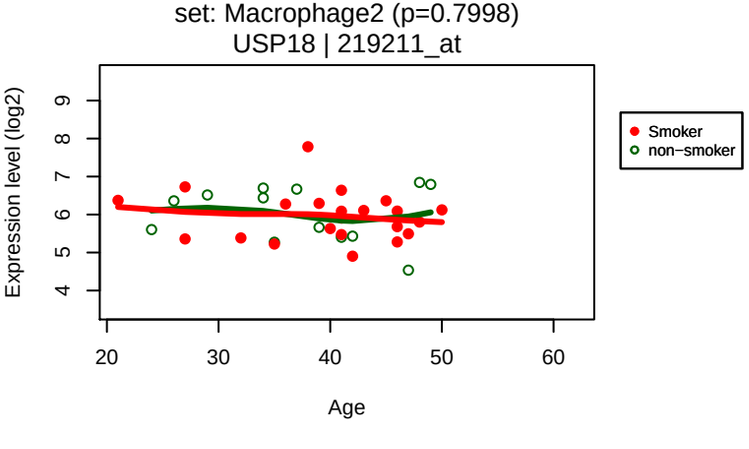

set: SmallAirway2 (p=0.5447)
USP18 | 219211_at
Expression level (log2)
Age
Smoker
non−smoker

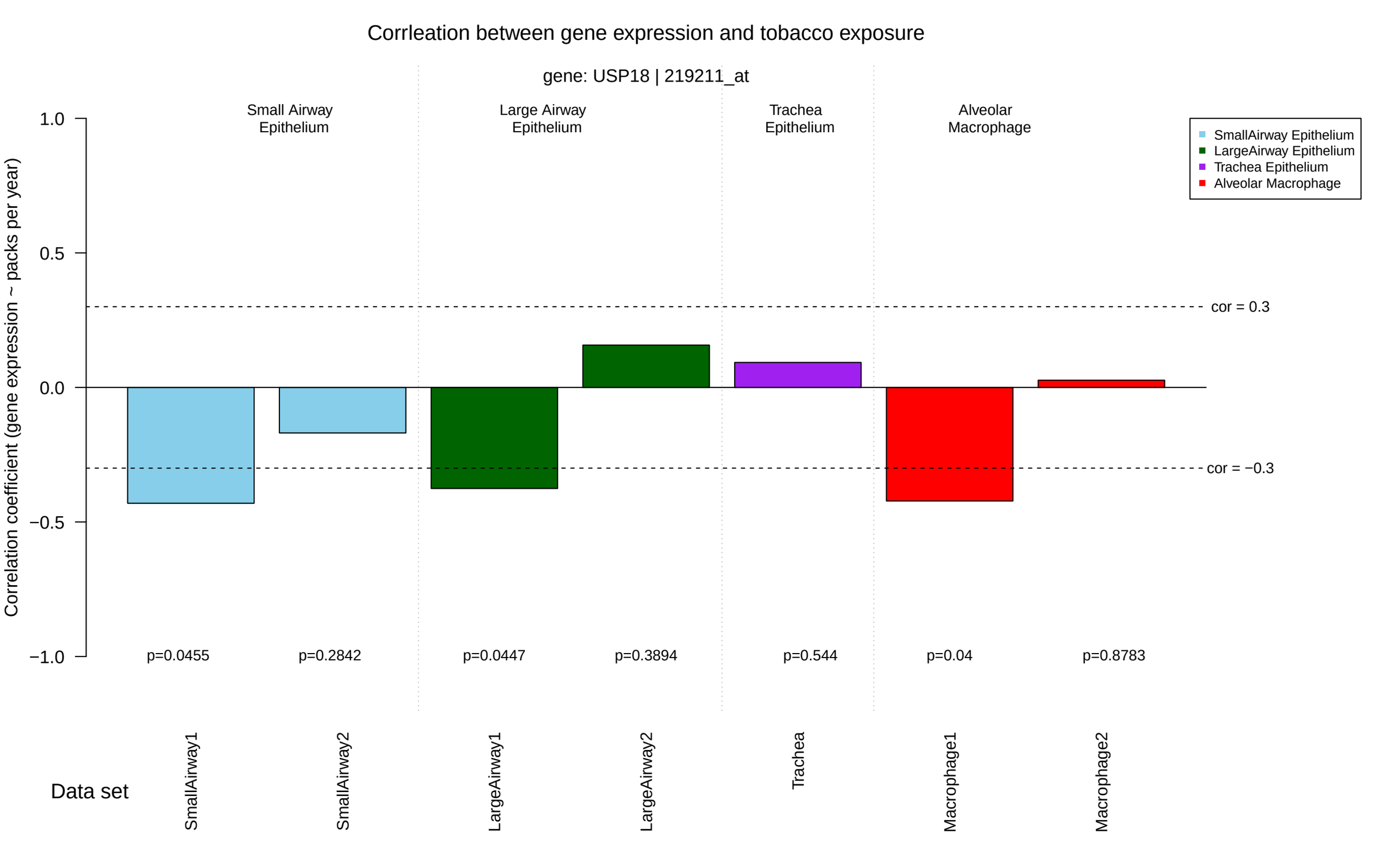

set: LargeAirway1 (p=0.0971)
USP18 | 219211_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2501)
USP18 | 219211_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7618)
USP18 | 219211_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0102)
USP18 | 219211_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7998)
USP18 | 219211_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP18 | 219211_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0455
p=0.2842
p=0.0447
p=0.3894
p=0.544
p=0.04
p=0.8783
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2